

Initiating Search

November 5, 2025, 8:34 PM

• Search:

CAS Lexicon Search:

Thioethers - Preferred Concept

Search Tasks

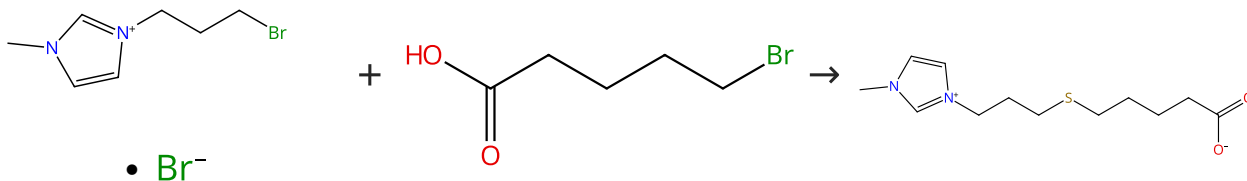
Task	Result Type	View
Returned Reference Results from CAS Lexicon (8,233)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (2,008)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2015 to 2025	

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (75)

31-026-CAS-5847544

Steps: 1 Yield: 100%

Modular Bidentate Hybrid NHC-Thioether Ligands for the Stabilization of Palladium Nanoparticles in Various Solvents

By: Ruehling, Andreas; et al

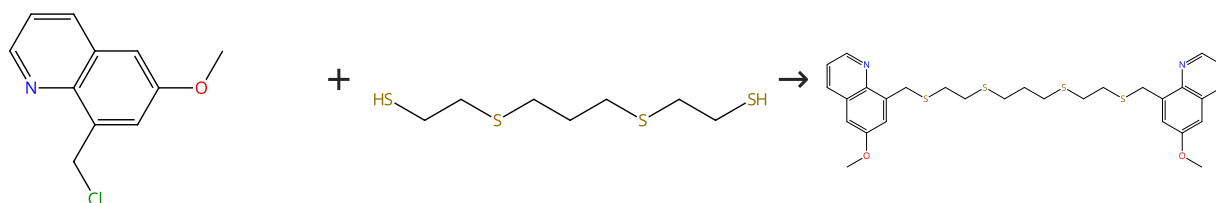
Angewandte Chemie, International Edition (2016), 55(19), 5856-5860.

1.1 Reagents: Potassium carbonate
Solvents: Ethanol; 3 d, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (37)

31-026-CAS-10598780

Steps: 1 Yield: 100%

Off-On, Ratiometric, and On-Off Fluorescence Responses of Thioether-Linked Bisquinolines toward Hg^{2+} and Fe^{3+} Ions

By: Mikata, Yuji; et al

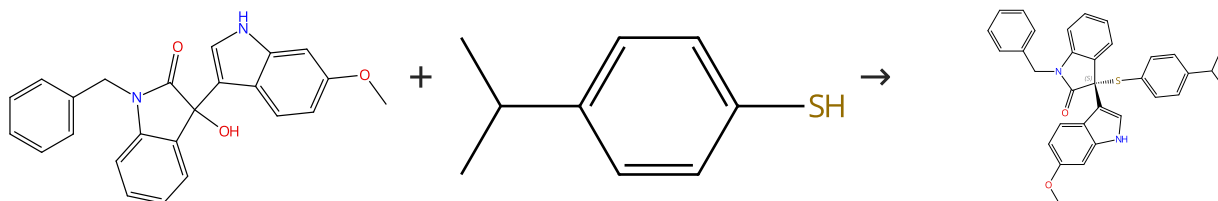
European Journal of Inorganic Chemistry (2015), 2015(22), 3769-3780.

1.1 Reagents: Potassium carbonate, Potassium iodide
Solvents: Acetonitrile; 2 d, rt $\rightarrow$ reflux

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (66)

Absolute stereochemistry shown,
Rotation (+)

31-614-CAS-24895262 Steps: 1 Yield: 100% 1.1 Catalysts: Dinaphtho[2,1-<i>d</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin-4-amine, 8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-<i>N</i>-[(11<i>bR</i>)-8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-4-oxidodiphospho[2,1-<i>d</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin-4-yl]-, 4-oxide, (11<i>bR</i>)- Solvents: Diethyl ether; 12 h, 10 °C Experimental Protocols	Imidodiphosphoric Acids Catalyzed Asymmetric Functionalization with Thiols: Access to Oxindole Derived-Chiral Thioethers By: Qin, Xiangshuo; et al Advanced Synthesis & Catalysis (2022), 364(1), 225-231.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



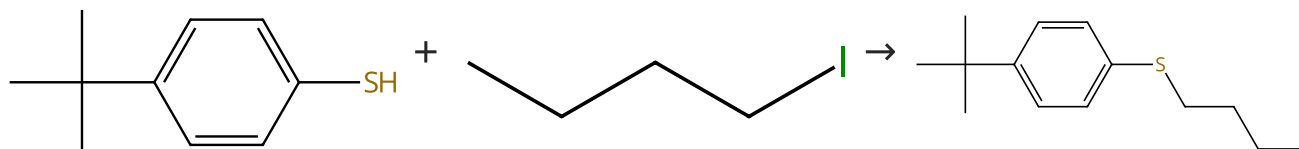
Suppliers (53)

Suppliers (188)

31-026-CAS-22112234 Steps: 1 Yield: 100% 1.1 Reagents: Hydrogen bromide Solvents: Acetic acid; 45 min, rt Experimental Protocols	An expeditious and efficient bromomethylation of thiols: enabling bromomethyl sulfides as useful building blocks By: Silva-Cuevas, Carolina; et al RSC Advances (2018), 8(43), 24654-24659.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (73)

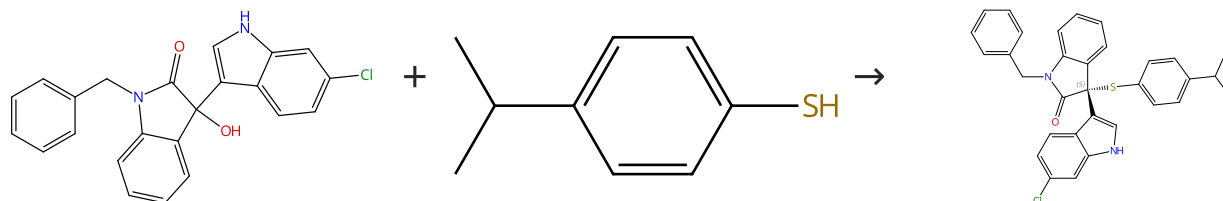
Suppliers (57)

Suppliers (7)

31-026-CAS-23901817 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Solvents: Methanol; 12 h, rt 1.2 Reagents: Water; rt Experimental Protocols	Thioether Ligand-Enabled Cationic Palladium(II)-Catalyzed Electrophilic C-H Arylation of α,β-Unsaturated Oxime Ethers By: Yamada, Takahiro; et al Journal of Organic Chemistry (2020), 85(19), 12315-12328.
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%

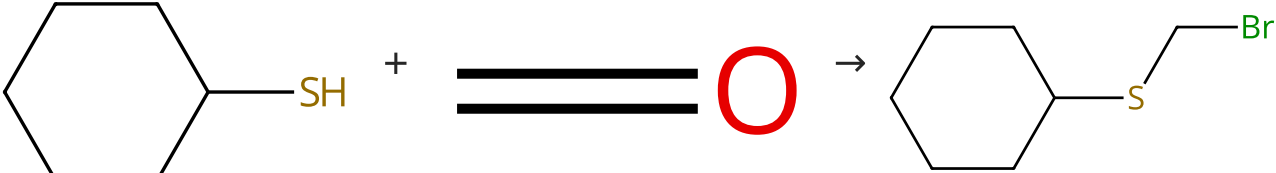


Suppliers (66)

Absolute stereochemistry shown,
Rotation (+)

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%


C1CCCCC1S + O=C=O >> C1CCCCC1SCCBr

 Suppliers (35)

 Suppliers (188)

31-026-CAS-21174915	Steps: 1 Yield: 100%	An expeditious and efficient bromomethylation of thiols: enabling bromomethyl sulfides as useful building blocks
1.1 Reagents: Hydrogen bromide Solvents: Acetic acid; 10 min, rt		By: Silva-Cuevas, Carolina; et al
Experimental Protocols		RSC Advances (2018), 8(43), 24654-24659.

Scheme 8 (1 Reaction) Steps: 1 Yield: 100%

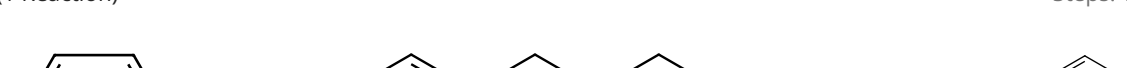

ClCC1=CC=C2C(=C1)N=CC=C2.CSCCSCCSCCSC
 $\rightarrow$
CSCCSCCSCCSCC1=CC=C2C(=C1)N=CC=C2

 Suppliers (16)

 Suppliers (37)

31-026-CAS-8188114	Steps: 1 Yield: 100%	Off-On, Ratiometric, and On-Off Fluorescence Responses of Thioether-Linked Bisquinolines toward Hg ²⁺ and Fe ³⁺ Ions
1.1 Reagents: Potassium carbonate, Potassium iodide Solvents: Acetonitrile; 2 d, rt → reflux		By: Mikata, Yuji; et al
Experimental Protocols		European Journal of Inorganic Chemistry (2015), 2015(22), 3769-3780.

Scheme 9 (1 Reaction) Steps: 1 Yield: 100%

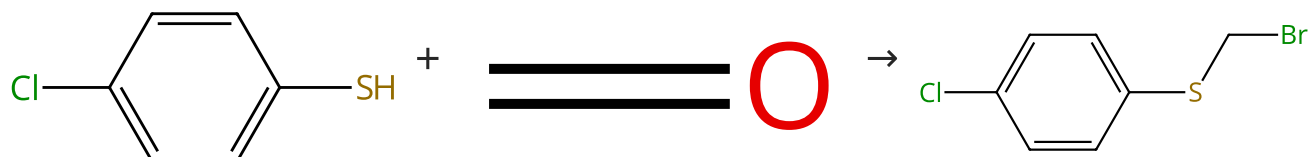

The reaction scheme shows the synthesis of 4-methoxy-1-phenylpropane-1-thiol. The starting materials are 4-mercaptophenyl methyl ether (a benzene ring with a methoxy group at position 1 and a thiol group at position 4) and 3-bromopropylbenzene (a benzene ring with a 3-bromopropyl group at position 1). The products are 4-methoxy-1-phenylpropane-1-thiol (a benzene ring with a methoxy group at position 1 and a 3-phenylpropylthio group at position 4) and hydrogen bromide (HBr). The reaction is catalyzed by potassium carbonate (K2CO3) in N-methyl-2-pyrrolidone (NMP) at 100 °C for 12 hours.

[Suppliers \(70\)](#) [Suppliers \(91\)](#)

31-614-CAS-45380027	Steps: 1 Yield: 100%	Photocatalytic and Chemoselective H/D Exchange at α-Thio C (sp^3)-H Bonds
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt; 1 h, 80 °C		By: Ogasahara, Riku; et al
Experimental Protocols		Journal of the American Chemical Society (2025), 147(18), 15499-15509.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (64)

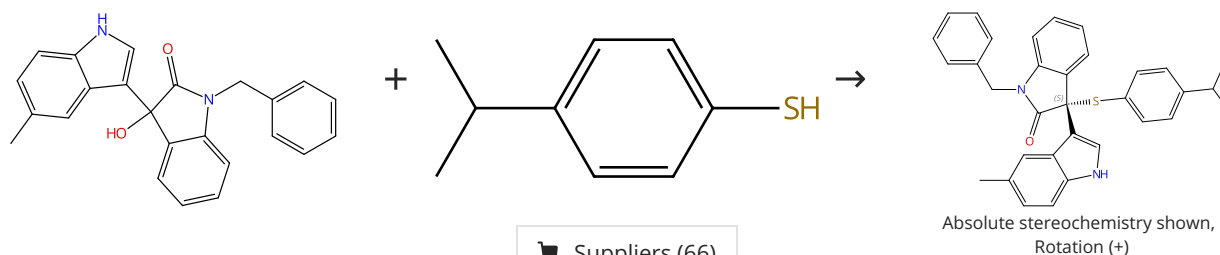
Suppliers (188)

Suppliers (2)

31-026-CAS-21054220	Steps: 1 Yield: 100%	An expeditious and efficient bromomethylation of thiols: enabling bromomethyl sulfides as useful building blocks
1.1 Reagents: Hydrogen bromide Solvents: Acetic acid; 45 min, rt		By: Silva-Cuevas, Carolina; et al
Experimental Protocols		RSC Advances (2018), 8(43), 24654-24659.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%

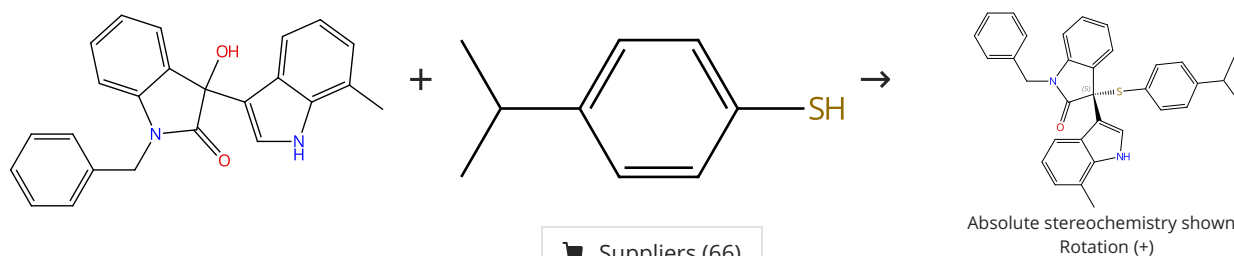


Suppliers (66)

31-614-CAS-24895266	Steps: 1 Yield: 100%	Imidodiphosphoric Acids Catalyzed Asymmetric Functionalization with Thiols: Access to Oxindole Derived-Chiral Thioethers
1.1 Catalysts: Dinaphtho[2,1- <i>d</i> :1',2'- <i>f</i>][1,3,2]dioxaphosphepin-4-amine, 8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl- <i>N</i> -[(11 <i>b</i> <i>R</i>)-8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-4-oxidodinaaphtho[2,1- <i>d</i> :1',2'- <i>f</i>][1,3,2]dioxaphosphepin-4-yl]-, 4-oxide, (11 <i>b</i> <i>R</i>)- Solvents: Diethyl ether; 12 h, 10 °C		By: Qin, Xiangshuo; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2022), 364(1), 225-231.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%

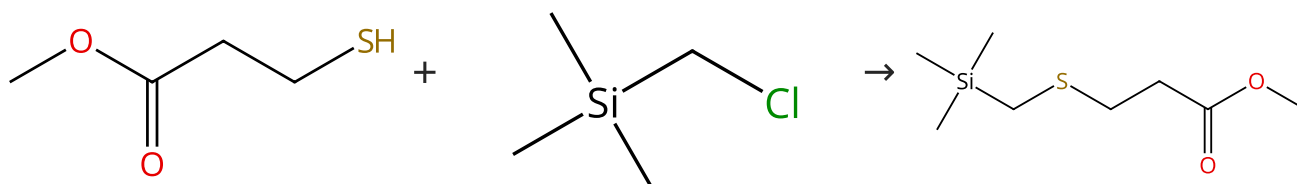


Suppliers (66)

31-614-CAS-24895248 Steps: 1 Yield: 100% 1.1 Catalysts: Dinaphtho[2,1-<i>d</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin-4-amine, 8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-<i>N</i>-[(11<i>b</i><i>R</i>)-8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-4-oxidodiphospho[2,1-<i>d</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin-4-yl]-, 4-oxide, (11<i>b</i><i>R</i>)- Solvents: Diethyl ether; 12 h, 10 °C Experimental Protocols	Imidodiphosphoric Acids Catalyzed Asymmetric Functionalization with Thiols: Access to Oxindole Derived-Chiral Thioethers By: Qin, Xiangshuo; et al Advanced Synthesis & Catalysis (2022), 364(1), 225-231.
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Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



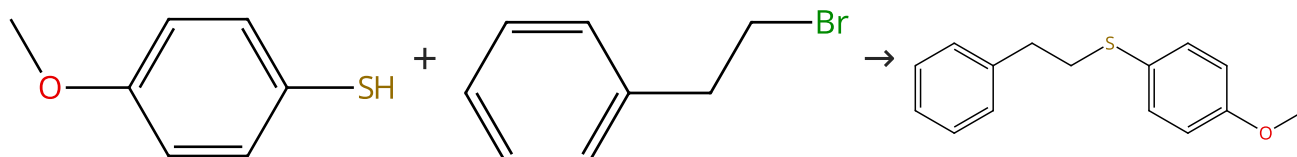
Suppliers (75)

Suppliers (61)

31-614-CAS-34764685 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 3 h, 70 °C Experimental Protocols	Chromium/Photoredox Dual Catalyzed Synthesis of α-Benzylic Alcohols, Isochromanones, 1,2-Oxy Alcohols and 1,2-Thio Alcohols By: Dutta, Subhabrata; et al Angewandte Chemie, International Edition (2022), 61(49), e202212136.
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Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

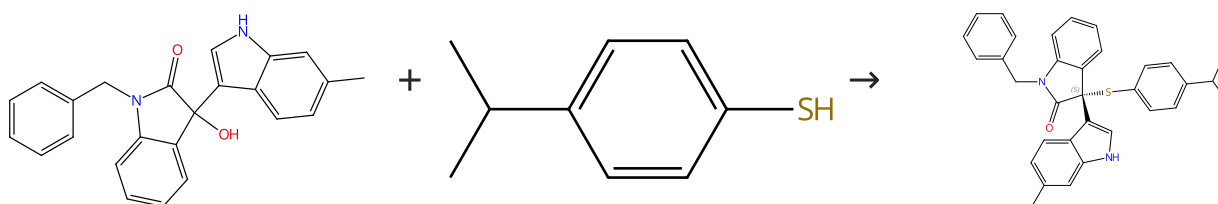
Suppliers (78)

Suppliers (2)

31-614-CAS-45380031 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt; 1.5 h, 80 °C Experimental Protocols	Photocatalytic and Chemoselective H/D Exchange at α-Thio C (sp^3)-H Bonds By: Ogasahara, Riku; et al Journal of the American Chemical Society (2025), 147(18), 15499-15509.
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Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



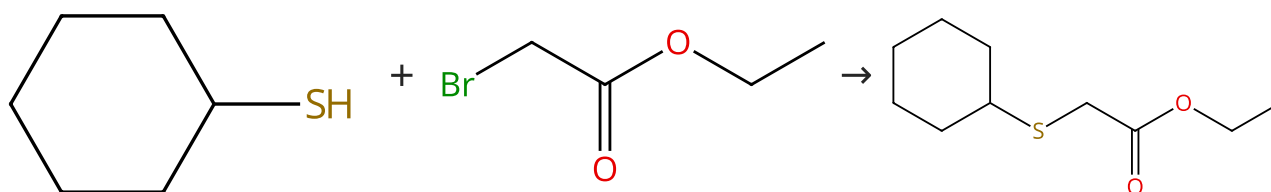
Suppliers (66)

Absolute stereochemistry shown,
Rotation (+)

31-614-CAS-24895263 Steps: 1 Yield: 100% 1.1 Catalysts: Dinaphtho[2,1-<i>d</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin-4-amine, 8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-<i>N</i>-[(11<i>bR</i>)-8,9,10,11,12,13,14,15-octahydro-2,6-di-1-naphthalenyl-4-oxidodiphospho[2,1-<i>d</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin-4-yl]-, 4-oxide, (11<i>bR</i>)- Solvents: Diethyl ether; 12 h, 10 °C Experimental Protocols	Imidodiphosphoric Acids Catalyzed Asymmetric Functionalization with Thiols: Access to Oxindole Derived-Chiral Thioethers By: Qin, Xiangshuo; et al Advanced Synthesis & Catalysis (2022), 364(1), 225-231.
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Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (35)

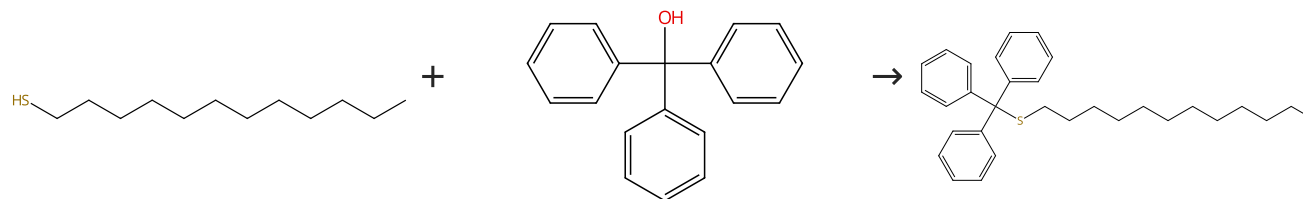
Suppliers (47)

Suppliers (4)

31-614-CAS-35470135 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetone; 30 min, rt; 6 h, rt Experimental Protocols	Novel S1R agonists counteracting NMDA excitotoxicity and oxidative stress: A step forward in the discovery of neuroprotective agents By: Linciano, Pasquale; et al European Journal of Medicinal Chemistry (2023), 249, 115163.
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Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

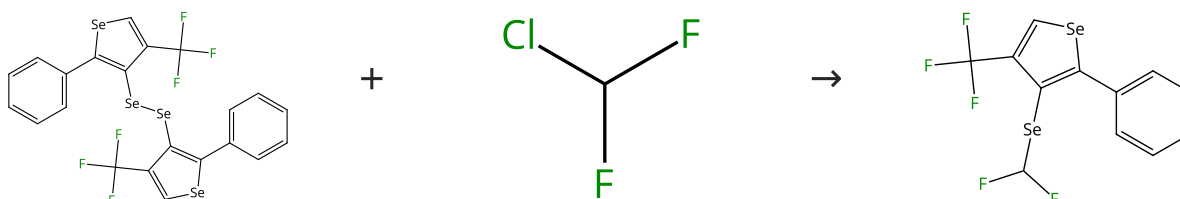
Suppliers (91)

Suppliers (2)

31-026-CAS-22218951 Steps: 1 Yield: 100% 1.1 Reagents: Trifluoroacetic acid Solvents: Chloroform; 3.5 h, rt Experimental Protocols	Functionalization of a Single C-F Bond of Trifluoromethylarenes Assisted by an ortho-Silyl Group Using a Trityl-Based All-in-One Reagent with Ytterbium Triflate Catalyst By: Kim, Youngchan; et al Chemistry - A European Journal (2020), 26(28), 6136-6140.
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Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%

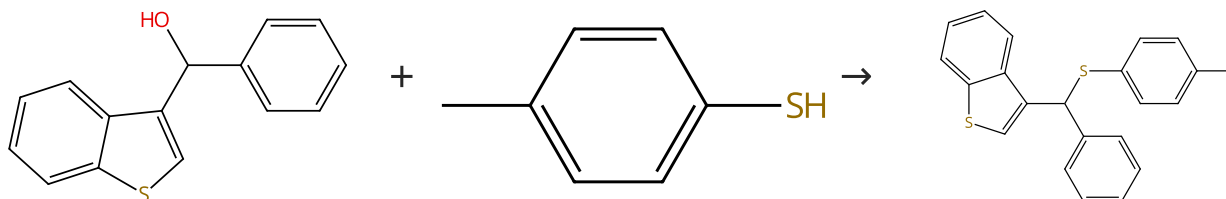


Suppliers (15)

31-026-CAS-21439911	Steps: 1 Yield: 100%	Precise Construction of SCF ₂ H or SeCF ₂ H Groups on Heteroarenes Generated in Situ from CF ₃ -Containing 1,3-Enynes
1.1 Reagents: Potassium carbonate, Water, S ₈ Solvents: Dimethylformamide; 10 h, 120 °C		By: Jin, Shengnan; et al
Experimental Protocols		Organic Letters (2020), 22(2), 615-619.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



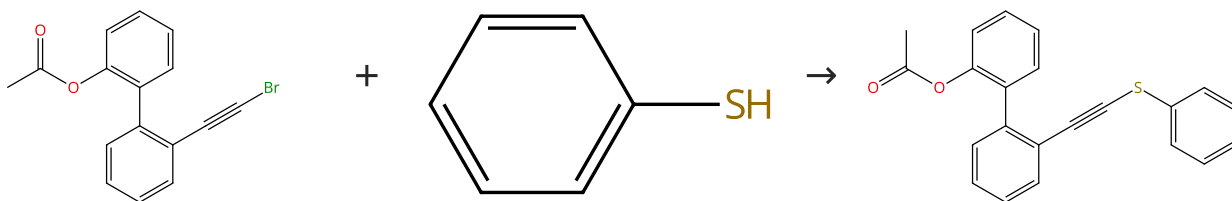
Suppliers (4)

Suppliers (56)

31-026-CAS-19903617	Steps: 1 Yield: 99%	B(C ₆ F ₅) ₃ catalyzed direct nucleophilic substitution of benzylic alcohols: an effective method of constructing C-O, C-S and C-C bonds from benzylic alcohols
1.1 Catalysts: Tris(pentafluorophenyl)borane Solvents: 1,2-Dichloroethane; 6 h, 60 °C		By: Meng, Shan-Shui; et al
		RSC Advances (2018), 8(54), 30946-30949.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%

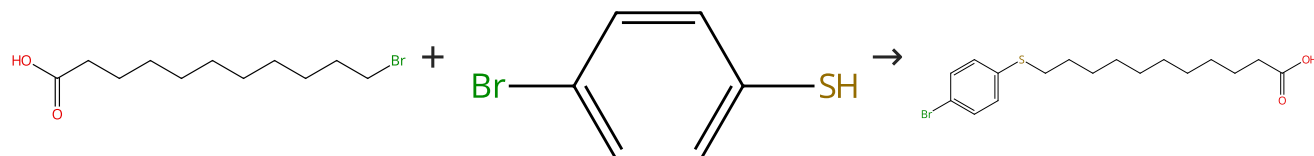


Suppliers (44)

31-614-CAS-38560662	Steps: 1 Yield: 99%	Chiral Bronsted acid-catalyzed asymmetric dearomative spirocyclization of alkynyl thioethers
1.1 Reagents: 2,6-Lutidine Catalysts: Tetrakis(acetonitrile)copper(1+) hexafluoro phosphate, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Acetonitrile; 10 - 60 min, rt		By: Fan, Xin-Yang; et al
Experimental Protocols		Green Chemistry (2023), 25(24), 10638-10643.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



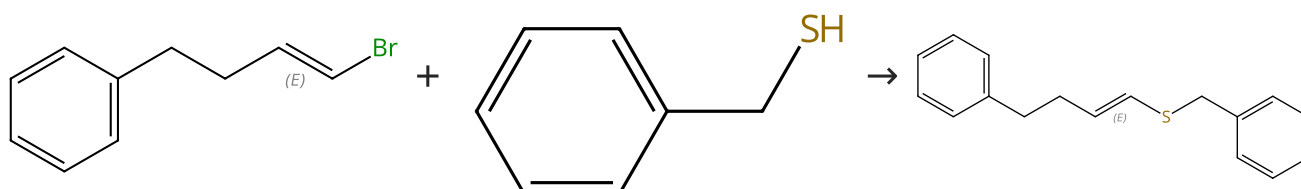
Suppliers (84)

Suppliers (77)

31-614-CAS-39002439	Steps: 1 Yield: 99%	Distinct twist-bend nematic phase behaviors associated with the ester-linkage direction of thioether-linked liquid crystal dimers
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 12 h, rt		By: Arakawa, Yuki; et al
Experimental Protocols		Materials Advances (2021), 2(1), 261-272.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (53)

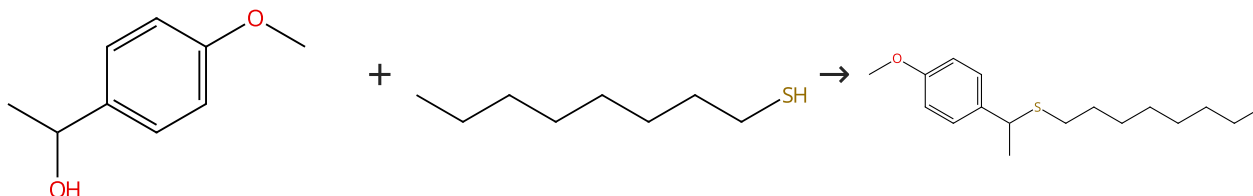
Double bond geometry shown

Suppliers (9)

31-614-CAS-36088385	Steps: 1 Yield: 99%	Alkenyl Thioetherification Enabled by Nickel Catalysis
1.1 Reagents: Sodium <i>tert</i> -butoxide, Zinc Catalysts: (<i>SP</i> -5-34)-Chloro[1,1'-(9,9-dimethyl-9 <i>H</i> -xanthene-4,5-diyl- κ <i>O</i>)bis[1,1-diphenylphosphine- κ <i>P</i>]](2-methylphenyl) nickel Solvents: Tetrahydrofuran; 12 h, 50 °C		By: Xu, Bing; et al
Experimental Protocols		Journal of Organic Chemistry (2023), 88(7), 4592-4605.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



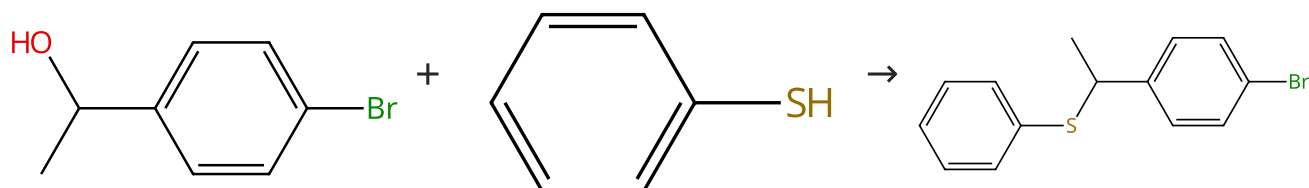
Suppliers (65)

Suppliers (60)

31-614-CAS-46386025	Steps: 1 Yield: 99%	Functionalization of Electron-Rich Secondary Benzyl Alcohols in HFIP: From Thioethers to Trisubstituted Methanes
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 20 h, 80 °C		By: Markwitz, Martyna; et al
Experimental Protocols		Journal of Organic Chemistry (2025), 90(27), 9513-9518.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



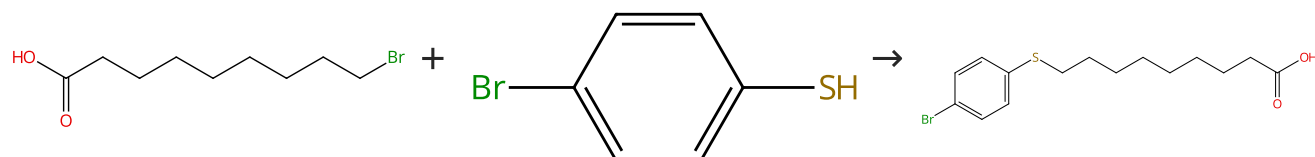
Suppliers (73)

Suppliers (44)

31-026-CAS-17876657	Steps: 1 Yield: 99%	Highly Efficient and Chemoselective Tertiary and Secondary Benzylation of Thiols Catalyzed by Indium (III) Triflate
1.1 Reagents: Oxygen Catalysts: Indium triflate Solvents: Nitromethane; < 30 min, 80 °C		By: Kucinski, Krzysztof; et al
Experimental Protocols		European Journal of Organic Chemistry (2017), 2017(37), 5572-5581.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



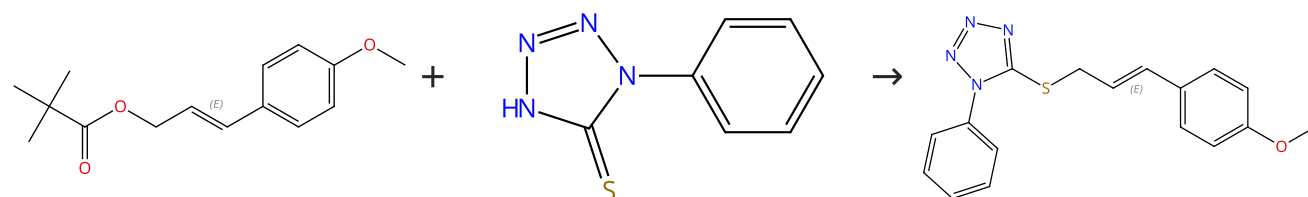
Suppliers (81)

Suppliers (77)

31-614-CAS-39002441	Steps: 1 Yield: 99%	Distinct twist-bend nematic phase behaviors associated with the ester-linkage direction of thioether-linked liquid crystal dimers
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 12 h, rt		By: Arakawa, Yuki; et al
Experimental Protocols		Materials Advances (2021), 2(1), 261-272.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (76)

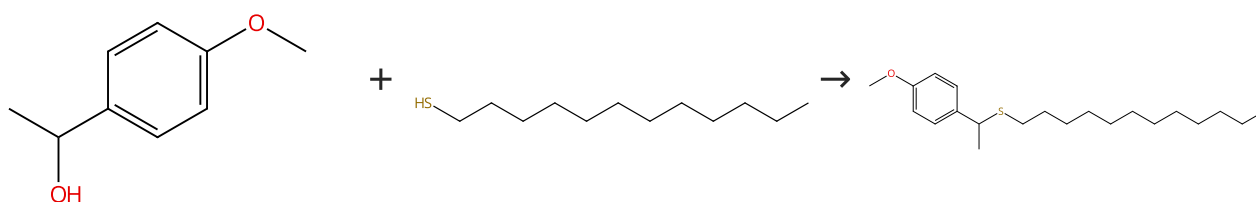
Double bond geometry shown

Supplier (1)

31-026-CAS-18293654	Steps: 1 Yield: 99%	Direct Synthesis of Thioethers from Carboxylates and Thiols Catalyzed by FeCl₃
1.1 Catalysts: Iron chloride (FeCl ₃) Solvents: Acetonitrile; 0.5 h, 80 °C		By: Venkatesham, Kunuru; et al
Experimental Protocols		Journal of Organic Chemistry (2015), 80(22), 11611-11617.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



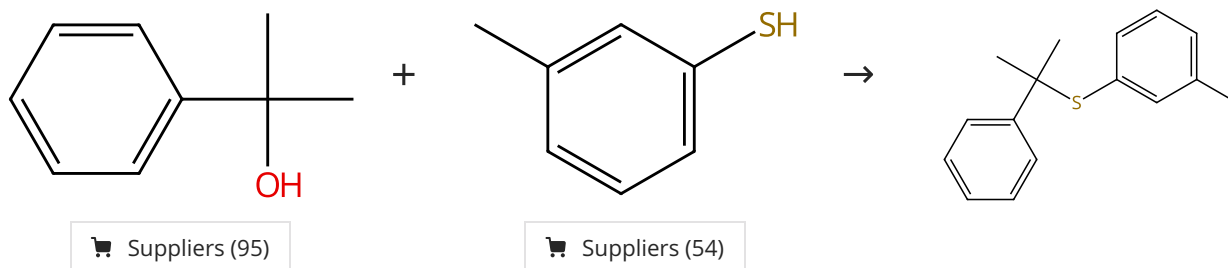
Suppliers (65)

Suppliers (57)

31-026-CAS-23893970	Steps: 1 Yield: 99%	Direct Nucleophilic Substitution of Alcohols Using an Immobilized Oxovanadium Catalyst
1.1 Catalysts: (7-4)-Oxotris(1,1,1-triphenylsilanolato)vanadium (silica supported) Solvents: Dichloromethane; 1 h, rt		By: Nishio, Tomoya; et al
Experimental Protocols		European Journal of Organic Chemistry (2021), 2021(31), 4417-4422.

Scheme 28 (1 Reaction)

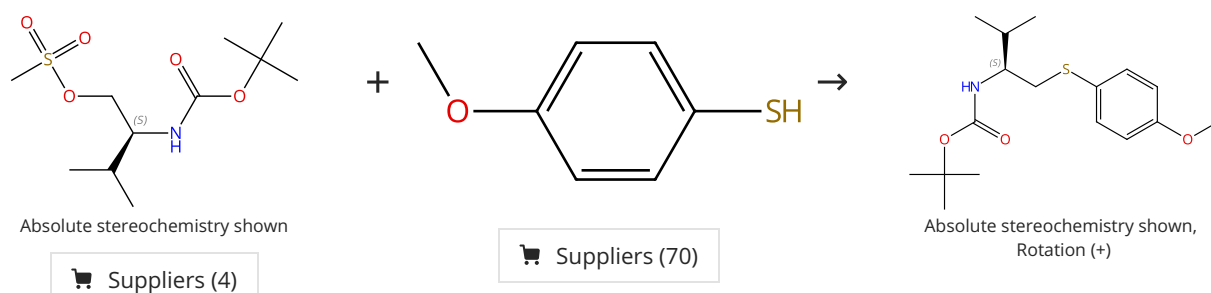
Steps: 1 Yield: 99%



31-026-CAS-17876617	Steps: 1 Yield: 99%	Highly Efficient and Chemoselective Tertiary and Secondary Benzylolation of Thiols Catalyzed by Indium (III) Triflate
1.1 Reagents: Oxygen Catalysts: Indium triflate Solvents: Nitromethane; < 30 min, rt		By: Kucinski, Krzysztof; et al
Experimental Protocols		European Journal of Organic Chemistry (2017), 2017(37), 5572-5581.

Scheme 29 (1 Reaction)

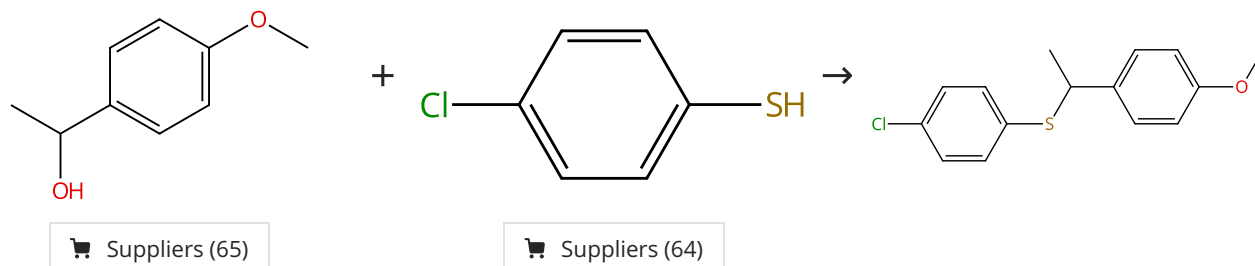
Steps: 1 Yield: 99%



31-614-CAS-24141811	Steps: 1 Yield: 99%	Electrochemical Deoxygenative Thiolation of Preactivated Alcohols and Ketones
1.1 Reagents: Tetrabutylammonium acetate Catalysts: 2,2'-Bipyridine, Bis(1,5-cyclooctadiene)nickel Solvents: Dimethylacetamide; 2 h, rt		By: Zhang, Feng; et al
Experimental Protocols		Organic Letters (2021), 23(19), 7524-7528.

Scheme 30 (1 Reaction)

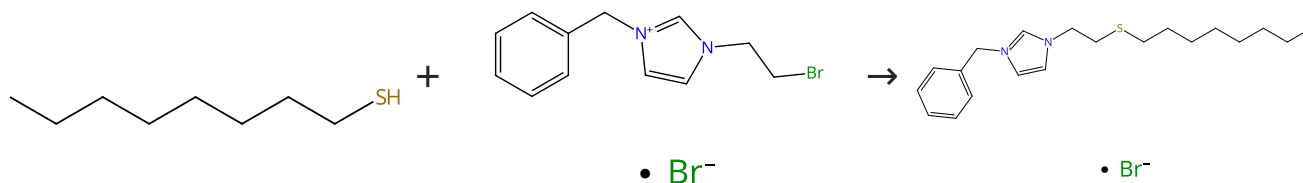
Steps: 1 Yield: 99%



31-614-CAS-46386017	Steps: 1 Yield: 99%	Functionalization of Electron-Rich Secondary Benzyl Alcohols in HFIP: From Thioethers to Trisubstituted Methanes
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 20 h, 80 °C		By: Markwitz, Martyna; et al
Experimental Protocols		Journal of Organic Chemistry (2025), 90(27), 9513-9518.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



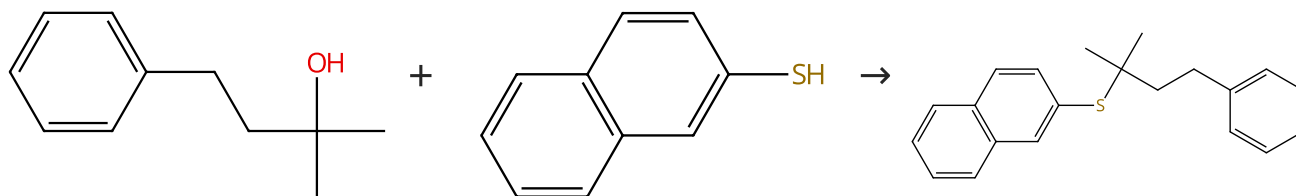
Suppliers (60)

Supplier (1)

31-026-CAS-21924902	Steps: 1 Yield: 99%	Synthesis, characterization, catalytic and biological application of half-sandwich ruthenium complexes bearing hemilabile (κ^2-C,S)-thioether-functionalised NHC ligands
1.1 Reagents: Sodium hydroxide Solvents: Acetonitrile, Water; 30 min, 25 °C		By: Chen, Weiguang; et al
1.2 15 - 48 h, 25 °C		Dalton Transactions (2020), 49(10), 3243-3252.
Experimental Protocols		

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



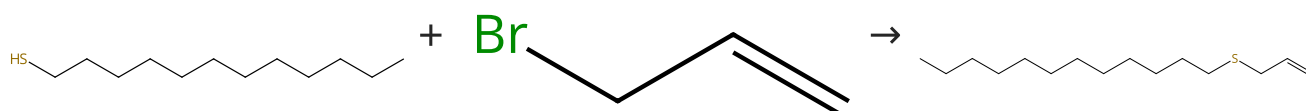
Suppliers (66)

Suppliers (69)

31-614-CAS-29439683	Steps: 1 Yield: 99%	Cu-catalyzed coupling of unactivated tertiary alkyl alcohols with thiols via C-O bond cleavage
1.1 Catalysts: Copper(II) triflate Solvents: 1,2-Dichloroethane; 12 h, 70 °C		By: Xu, Bing; et al
Experimental Protocols		Tetrahedron Letters (2022), 89, 153604.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (57)

Suppliers (57)

Suppliers (4)

Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%

The reaction scheme shows 1-indanol (a benzene ring fused to a five-membered ring with an -OH group) reacting with benzyl mercaptane (a benzene ring with a -CH₂-SH group). The product is 1-benzyl-1H-indole, where the -OH group of 1-indanol is replaced by a -CH₂-C₆H₅ group.

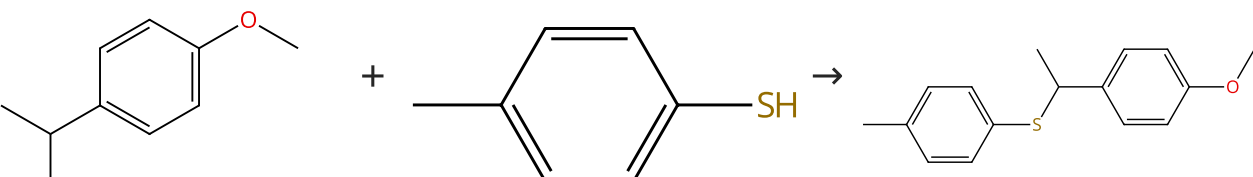
Suppliers (88)

Suppliers (44)

Suppliers (3)

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%


The reaction scheme shows 4-methoxyphenol (a benzene ring with a methoxy group at the para position and a hydroxyl group at the other para position) reacting with 4-mercaptobenzotoluene (a benzene ring with a methyl group at the para position and a thiol group at the other para position). The product is 4-methoxy-1-phenylpropane-1-thiol, where the thiol group of the starting material has been replaced by a 4-methoxyphenyl group.

 Suppliers (65)

 Suppliers (56)

31-026-CAS-19903619	Steps: 1 Yield: 99%	B(C₆F₅)₃ catalyzed direct nucleophilic substitution of benzylic alcohols: an effective method of constructing C-O, C-S and C-C bonds from benzylic alcohols By: Meng, Shan-Shui; et al RSC Advances (2018), 8(54), 30946-30949.
1.1 Catalysts: Tris(pentafluorophenyl)borane Solvents: 1,2-Dichloroethane; 6 h, 60 °C		

Scheme 36 (1 Reaction) Steps: 1 Yield: 99%


The reaction scheme shows the synthesis of 2-(benzylthio)propane-2-thiol. The starting materials are 2-(thiophen-2-yl)propan-2-ol (a thienyl-substituted tertiary alcohol) and benzylmercaptane (a primary thiol). The reaction proceeds via a nucleophilic substitution (S_N2) mechanism, where the sulfur atom of the thiol attacks the carbon bearing the hydroxyl group, displacing the hydroxide ion. The product is 2-(benzylthio)propane-2-thiol, which contains both a thioether and a thiol functional group.

CC(C)(O)c1ccsc1 + SCc1ccccc1 → SC(C)(SCc1ccccc1)c2ccsc2

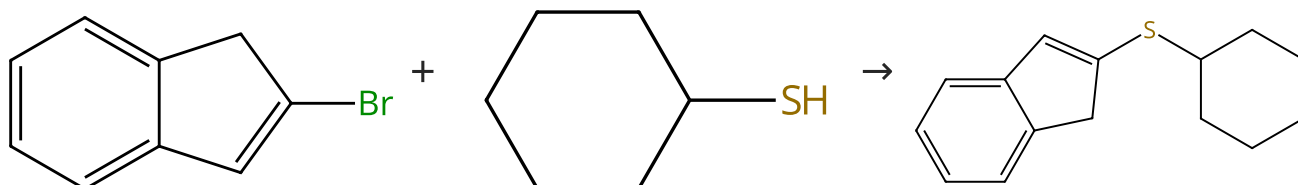
 Suppliers (49)

 Suppliers (53)

31-614-CAS-44293473	Steps: 1 Yield: 99%	Rapid C-S Coupling in Water via Ion-Pair-Catalyzed Dehydration
1.1 Catalysts: Triphenylcarbenium tetrakis(pentafluorophenyl) borate Solvents: Water; 5 min, 37 °C		By: Zhu, Yufei; et al Organic Letters (2025), 27(9), 2110-2115.
Experimental Protocols		

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



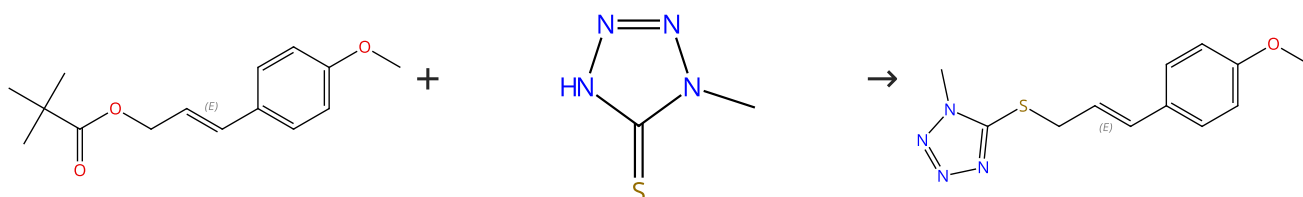
Suppliers (82)

Suppliers (35)

31-614-CAS-36088362	Steps: 1 Yield: 99%	Alkenyl Thioetherification Enabled by Nickel Catalysis
1.1 Reagents: Potassium acetate, Zinc Catalysts: (<i>SP</i> -5-34)-Chloro[1,1'-(9,9-dimethyl-9 <i>H</i> -xanthene-4,5-diyl-κ <i>O</i>)bis[1,1-diphenylphosphine-κ <i>P</i>]](2-methylphenyl) nickel Solvents: Tetrahydrofuran; 12 h, 50 °C		By: Xu, Bing; et al Journal of Organic Chemistry (2023), 88(7), 4592-4605.
Experimental Protocols		

Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (80)

Double bond geometry shown

Supplier (1)

31-026-CAS-18237853	Steps: 1 Yield: 99%	Direct Synthesis of Thioethers from Carboxylates and Thiols Catalyzed by FeCl₃
1.1 Catalysts: Iron chloride (FeCl ₃) Solvents: Acetonitrile; 0.5 h, 80 °C		By: Venkatesham, Kunuru; et al Journal of Organic Chemistry (2015), 80(22), 11611-11617.
Experimental Protocols		

Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (99)

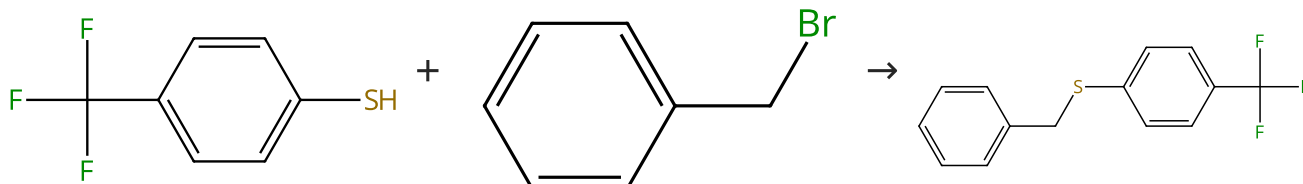
Suppliers (50)

Supplier (1)

31-014-CAS-22407561	Steps: 1 Yield: 99%	Synthesis of functional 1,2-dithiolanes from 1,3-bis-tert-butyl thioethers
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 3 h, rt		By: Scheutz, Georg M.; et al
1.2 Reagents: Sodium hydroxide Solvents: Diethyl ether, Water; rt		Organic & Biomolecular Chemistry (2020), 18(33), 6509-6513.
Experimental Protocols		

Scheme 40 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (66)

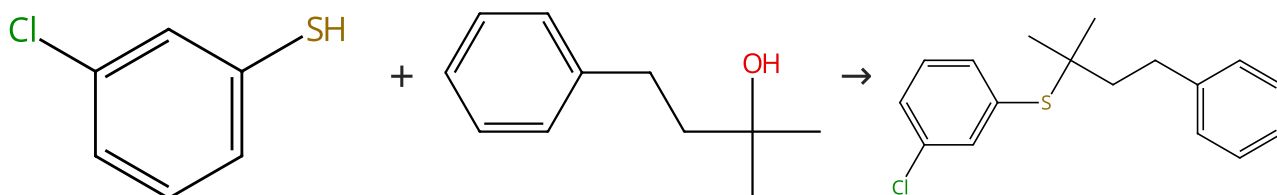
Suppliers (67)

Suppliers (5)

31-614-CAS-37288747	Steps: 1 Yield: 99%	Electroreductive Desulfurative Transformations with Thioethers as Alkyl Radical Precursors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 5 min, rt		By: Kuzmin, Julius; et al
1.2 overnight, 90 °C		Angewandte Chemie, International Edition (2023), 62(39), e202304272.
Experimental Protocols		

Scheme 41 (1 Reaction)

Steps: 1 Yield: 99%



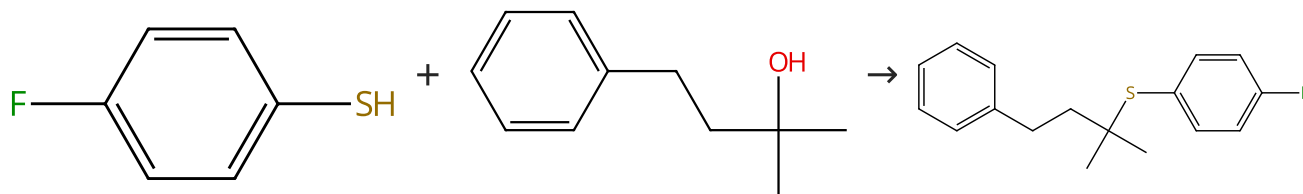
Suppliers (57)

Suppliers (66)

31-614-CAS-29439664	Steps: 1 Yield: 99%	Cu-catalyzed coupling of unactivated tertiary alkyl alcohols with thiols via C-O bond cleavage
1.1 Catalysts: Copper(II) triflate Solvents: 1,2-Dichloroethane; 12 h, 70 °C		By: Xu, Bing; et al
Experimental Protocols		Tetrahedron Letters (2022), 89, 153604.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 99%



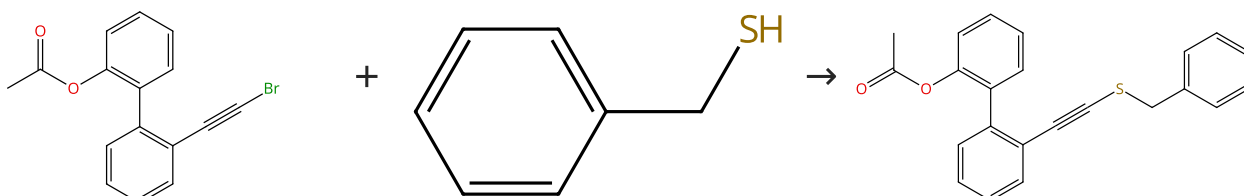
Suppliers (78)

Suppliers (66)

31-614-CAS-29439654	Steps: 1 Yield: 99%	Cu-catalyzed coupling of unactivated tertiary alkyl alcohols with thiols via C-O bond cleavage
1.1 Catalysts: Copper(II) triflate Solvents: 1,2-Dichloroethane; 12 h, 70 °C		By: Xu, Bing; et al
Experimental Protocols		Tetrahedron Letters (2022), 89, 153604.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%

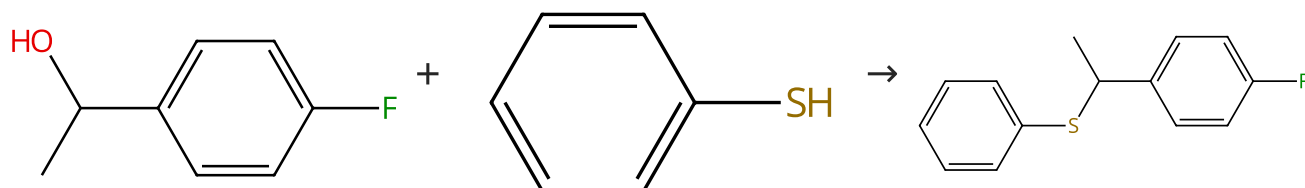


Suppliers (53)

31-614-CAS-38560657	Steps: 1 Yield: 99%	Chiral Bronsted acid-catalyzed asymmetric dearomative spirocyclization of alkynyl thioethers
1.1 Reagents: 2,6-Lutidine Catalysts: Tetrakis(acetonitrile)copper(1+) hexafluoro phosphate, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Acetonitrile; 10 - 60 min, rt		By: Fan, Xin-Yang; et al
Experimental Protocols		Green Chemistry (2023), 25(24), 10638-10643.

Scheme 44 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (76)

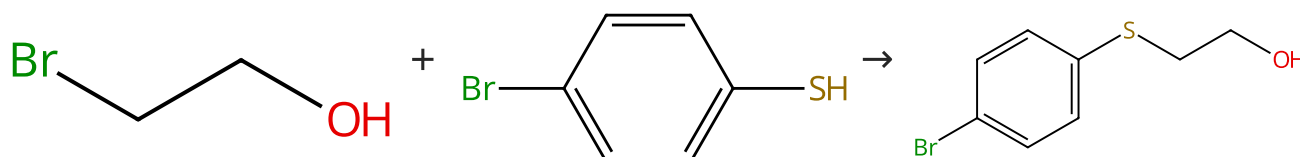
Suppliers (44)

Supplier (1)

31-026-CAS-17876654	Steps: 1 Yield: 99%	Highly Efficient and Chemoselective Tertiary and Secondary Benzoylation of Thiols Catalyzed by Indium (III) Triflate
1.1 Reagents: Oxygen Catalysts: Indium triflate Solvents: Nitromethane; < 30 min, 80 °C		By: Kucinski, Krzysztof; et al
Experimental Protocols		European Journal of Organic Chemistry (2017), 2017(37), 5572-5581.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (76)

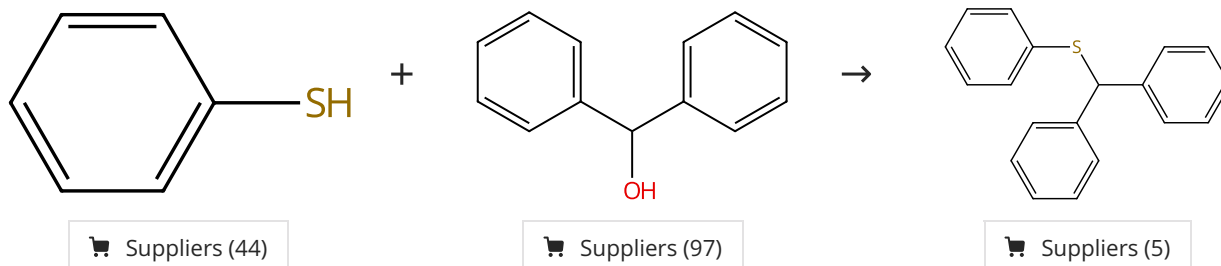
Suppliers (77)

Suppliers (34)

31-614-CAS-39002421	Steps: 1 Yield: 99%	Distinct twist-bend nematic phase behaviors associated with the ester-linkage direction of thioether-linked liquid crystal dimers
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 12 h, rt		By: Arakawa, Yuki; et al
Experimental Protocols		Materials Advances (2021), 2(1), 261-272.

Scheme 46 (2 Reactions)

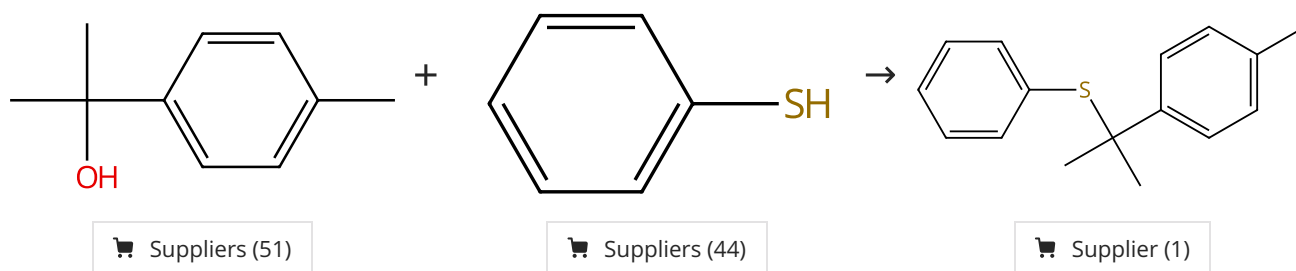
Steps: 1 Yield: 99%



31-614-CAS-38727756	Steps: 1 Yield: 99%	Access to thioethers from thiols and alcohols via homogeneous and heterogeneous catalysis
1.1 Catalysts: Trifluoromethanesulfonic acid Solvents: Nitromethane; rt; 2 h, 80 °C		By: Markwitz, Martyna; et al
1.2 Reagents: Potassium carbonate Catalysts: Trifluoromethanesulfonic acid; neutralized		Scientific Reports (2023), 13(1), 20624.
Experimental Protocols		
31-026-CAS-17876650	Steps: 1 Yield: 99%	Highly Efficient and Chemoselective Tertiary and Secondary Benzylation of Thiols Catalyzed by Indium (III) Triflate
1.1 Reagents: Oxygen Catalysts: Indium triflate Solvents: Nitromethane; < 30 min, 80 °C		By: Kucinski, Krzysztof; et al
Experimental Protocols		European Journal of Organic Chemistry (2017), 2017(37), 5572-5581.

Scheme 47 (1 Reaction)

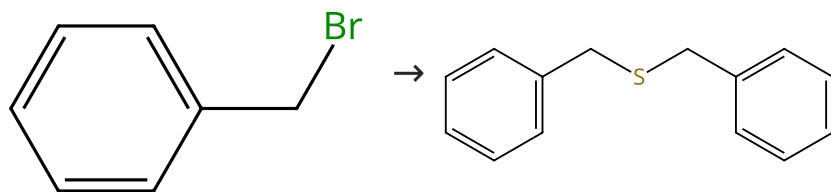
Steps: 1 Yield: 99%



31-026-CAS-17876628	Steps: 1 Yield: 99%	Highly Efficient and Chemoselective Tertiary and Secondary Benzylation of Thiols Catalyzed by Indium (III) Triflate
1.1 Reagents: Oxygen Catalysts: Indium triflate Solvents: Nitromethane; < 30 min, rt		By: Kucinski, Krzysztof; et al
Experimental Protocols		European Journal of Organic Chemistry (2017), 2017(37), 5572-5581.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (67)

Suppliers (57)

31-026-CAS-16138719

Steps: 1 Yield: 99%

Synthesis of Thioethers Using Triethylamine-Activated Phosphorus Decasulfide (P₄S₁₀)
1.1 Reagents: Triethylamine, Phosphorus sulfide (P₂S₅); 30 min, rt

1.2 20 h, rt

1.3 Reagents: Ammonium chloride

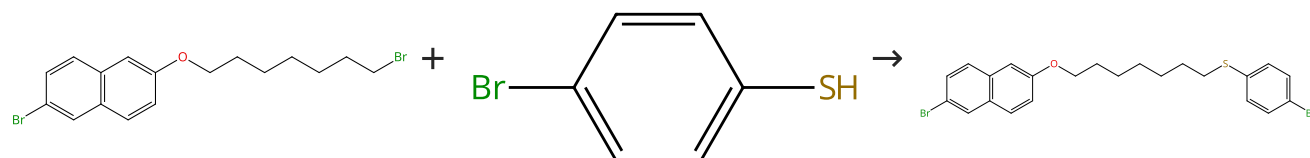
Solvents: Water; rt

By: Turkoglu, Gulsen; et al

Synthesis (2016), 48(20), 3618-3624.

Scheme 49 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

31-026-CAS-21930805

Steps: 1 Yield: 99%

Ether- and Thioether-Linked Naphthalene-Based Liquid-Crystal Dimers: Influence of Chalcogen Linkage and Mesogenic-Arm Symmetry on the Incidence and Stability of the Twist-Bend Nematic Phase

1.1 Reagents: Potassium carbonate

Catalysts: 18-Crown-6

Solvents: Dimethylformamide; 12 h, reflux

By: Arakawa, Yuki; et al

Chemistry - A European Journal (2020), 26(17), 3767-3775.

Experimental Protocols